

Semiconductor packaging industry update

EQUITY: JAPAN ELECTRONIC PARTS

Packaging substrate shipments up 36% y-y

Fiercer competition between TSMC and Intel in quad-die packaging

Packaging-related monthly statistics: May shipments hit record high after slowdown in April

According to the Current Survey of Production for May, released on 14 July, shipments of packaging substrates (rigid module substrates) rose 36% y-y to ¥27.8bn, surpassing the previous record high of ¥27.3bn set in March. Shipments in terms of m² rose 10% y-y and the average price per m² rose 23% to ¥1.356mn, above the previous record high of ¥1.226mn set in April. We infer that shipments of Rubin packages will pick up in earnest this summer (around August at the latest), and think shipments of these packages have been rising gradually.

Intel releases technical paper on EMIB-T at ECTC, we see evidence of improved power supply performance

There is a growing view in the packaging industry that the structure of the Rubin Ultra will be changed from four dies, which were announced at GTC in March, to two (see, for example, our [Quick Note - SemiAnalysis report on Rubin Ultra](#)), and some observers are starting to take the view that the Feynman will have a two-module structure with two dies in 3D stacking configuration. We think it may have become technically more difficult to combine larger interposers (greater than 5.5x the reticle size) with HBM4, which has more wiring, and we think companies developing cutting-edge packages using interposers (such as TSMC) may consider their future roadmap to be based on the use of 3.5D packaging, which combines 3D stacking with 2.5D packaging, until visibility improves on the shift to larger interposers.

Intel's Embedded Multi-die Interconnect Bridge Through-Silicon Via (EMIB-T) is 2.5D packaging that does not use interposers, and we expect the company to complete preparations for mass production of a product with 8x the reticle size, which will feature four dies, in 2026. The package substrate is 120x120mm, and it can integrate up to 12 HBM4. We think there have been moves towards uptake at Google, which is developing accelerators with quad-die structure. The technical paper "Scaling the EMIB-T Advanced Packaging Technology to Address the Future HPC/AI Demand" published by Intel engineers at ECTC in Florida in May provides details about EMIB-T silicon bridges. The paper says that in this technology, TSV (not found in previous EMIB models) is added to a 3x18mm silicon bridge in order to strengthen power supply, and a power mesh layer (power supply wiring layer) and MIM capacitor (using two metal layers to form a capacitor) are provided, resulting in 68–80% improvement in voltage reduction in the power supply network within the package as compared to EMIB. Put another way, the decline in power supply was too sharp for EMIB to be used in accelerator packages, and Intel appears to have overcome the power supply issues by means of the TSV and MIM capacitors used in EMIB-T.

We think TSMC has a strong competitive edge in power supply and heat management within package substrates, as it has leveraged the 3DFabric Alliance to develop products in collaboration with other companies, and we expect Intel to build a similar ecosystem and catch up rapidly. Core technologies that determine the strength of an ecosystem include power supply, heat management, co-packaged optics (CPO), and 3D stacking (hybrid bonding), and we are interested to see how the two companies build and strengthen their ecosystems in these areas.

Research Analysts

Japan electronic parts

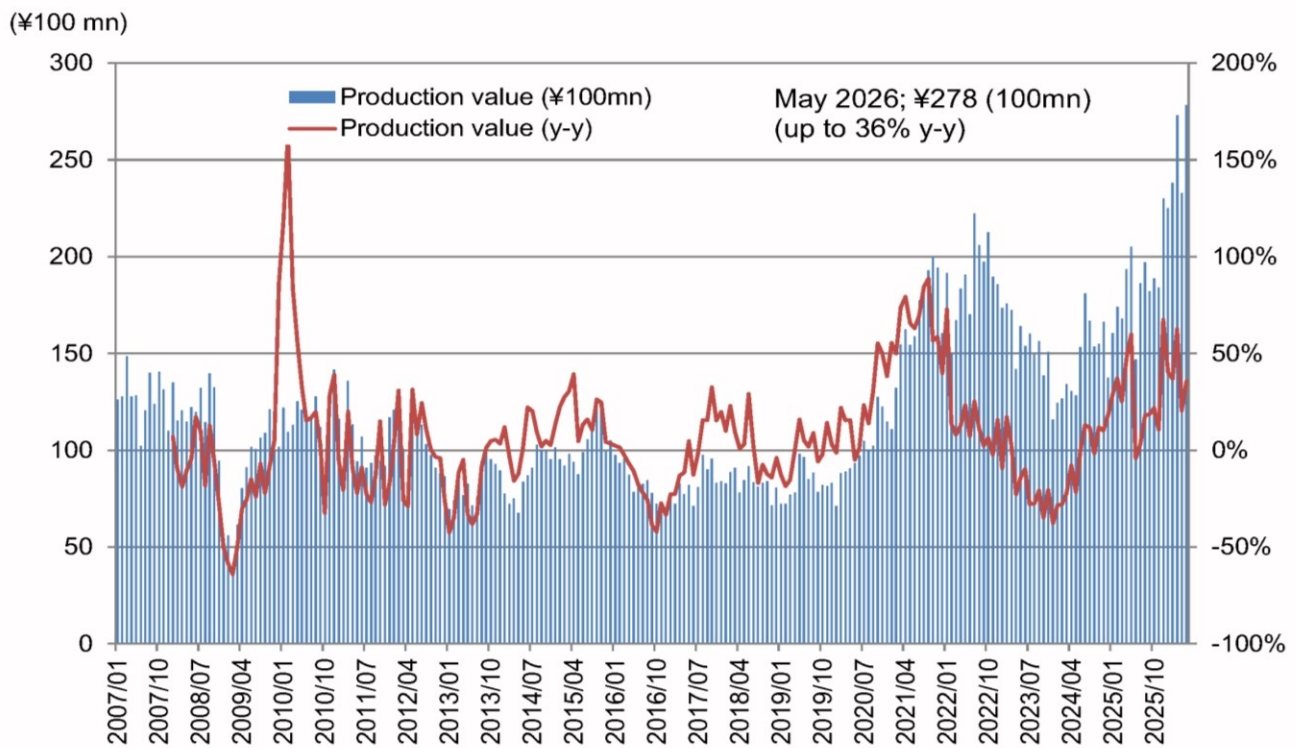
Manabu Akizuki - NSC

manabu.akizuki@nomura.com

+81 3 6703 1185

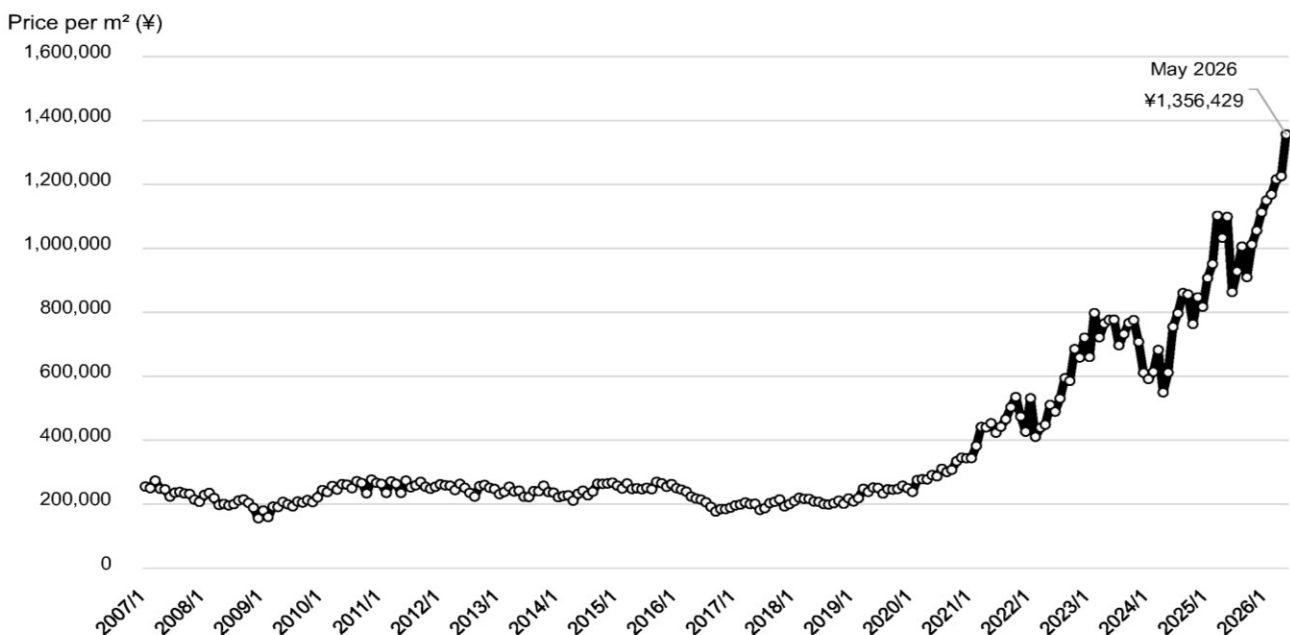
Reference figures

Fig. 1: Rigid PCB production value



Source: Nomura, based on Ministry of Economy, Trade & Industry's Current Survey of Production

Fig. 2: Price per m² for rigid module boards



Note: Module boards have components mounted on the board surface.

Source: Nomura, based on Ministry of Economy, Trade & Industry's Current Survey of Production

Appendix A-1

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Transactions involving foreign equities are subject to a domestic sales commission of up to 1.045% (tax included) of the transaction amount (which equals the local transaction amount plus local fees and taxes in the case of a purchase or the local transaction amount minus local fees and taxes in the case of a sale) (for transaction amounts of ¥750,000 and below, maximum domestic sales commission is ¥7,810 (tax included)). Local fees and taxes in foreign financial instruments markets vary by country/territory. When foreign equities are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Foreign equities carry the risk of losses owing to factors such as price fluctuations and foreign exchange rate fluctuations.

Margin transactions are subject to a sales commission of up to 1.43% (tax included) of the transaction amount (or a commission of ¥2,860 (tax included) for transactions of ¥200,000 or less), as well as management fees and rights handling fees. In addition, long margin transactions are subject to interest on the purchase amount, while short margin transactions are subject to fees for the lending of the shares borrowed. A margin equal to at least 30% of the transaction amount (at least 33% for online transactions) and at least ¥300,000 is required. With margin transactions, an amount up to roughly 3.3x the margin (roughly 3x for online transactions) may be traded. Margin transactions therefore carry the risk of losses in excess of the margin owing to share price fluctuations. For details, please thoroughly read the written materials provided, such as listed securities documents or documents delivered before making a contract.

Transactions involving convertible bonds are subject to a sales commission of up to 1.10% (tax included) of the transaction amount (or a commission of ¥4,400 (tax included) if this would be less than ¥4,400). When convertible bonds are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Convertible bonds carry the risk of losses owing to factors such as interest rate fluctuations and price fluctuations in the underlying stock. In addition, convertible bonds denominated in foreign currencies also carry the risk of losses owing to factors such as foreign exchange rate fluctuations.

When bonds are purchased via public offerings, secondary distributions, or other OTC transactions with Nomura Securities, only the purchase price shall be paid, with no sales commission charged. Bonds carry the risk of losses, as prices fluctuate in line with changes in market interest rates. Bond prices may also fall below the invested principal as a result of such factors as changes in the management and financial circumstances of the issuer, or changes in third-party valuations of the bond in question. In addition, foreign currency-denominated bonds also carry the risk of losses owing to factors such as foreign exchange rate fluctuations.

When Japanese government bonds (JGBs) for individual investors are purchased via public offerings, only the purchase price shall be paid, with no sales commission charged. As a rule, JGBs for individual investors may not be sold in the first 12 months after issuance. When JGBs for individual investors are sold before maturity, an amount calculated via the following formula will be subtracted from the par value of the bond plus accrued interest: (1) for 10-year variable rate bonds, an amount equal to the two preceding coupon payments (before tax) x 0.79685 will be used, (2) for 5-year and 3-year fixed rate bonds, an amount equal to the two preceding coupon payments (before tax) x 0.79685 will be used. When inflation-indexed JGBs are purchased via public offerings, secondary distributions (uridashi deals), or other OTC transactions with Nomura Securities, only the purchase price shall be paid, with no sales commission charged. Inflation-indexed JGBs carry the risk of losses, as prices fluctuate in line with changes in market interest rates and fluctuations in the nationwide consumer price index. The notional principal of inflation-indexed JGBs changes in line with the rate of change in nationwide CPI inflation from the time of its issuance. The amount of the coupon payment is calculated by multiplying the coupon rate by the notional principal at the time of payment. The maturity value is the amount of the notional principal when the issue becomes due. For J117 and subsequent issues, the maturity value shall not undercut the face amount. Purchases of investment trusts (and sales of some investment trusts) are subject to a purchase or sales fee of up to 5.5% (tax included) of the transaction amount. Also, a direct cost that may be incurred when selling investment trusts is a fee of up to 2.0% of the unit price at the time of redemption. Indirect costs that may be incurred during the course of holding investment trusts include, for domestic investment trusts, an asset

management fee (trust fee) of up to 5.5% (tax included/annualized basis) of the net assets in trust, as well as fees based on investment performance. Other indirect costs may also be incurred. For foreign investment trusts, indirect fees may be incurred during the course of holding such as investment company compensation.

Investment trusts invest mainly in securities such as Japanese and foreign equities and bonds, whose prices fluctuate. Investment trust unit prices fluctuate owing to price fluctuations in the underlying assets and to foreign exchange rate fluctuations. As such, investment trusts carry the risk of losses. Fees and risks vary by investment trust. Maximum applicable fees are subject to change; please thoroughly read the written materials provided, such as prospectuses or documents delivered before making a contract.

In interest rate swap transactions and USD/JPY basis swap transactions ("interest rate swap transactions, etc."), only the agreed transaction payments shall be made on the settlement dates. Some interest rate swap transactions, etc. may require pledging of margin collateral. In some of these cases, transaction payments may exceed the amount of collateral. There shall be no advance notification of required collateral value or collateral ratios as they vary depending on the transaction. Interest rate swap transactions, etc. carry the risk of losses owing to fluctuations in market prices in the interest rate, currency and other markets, as well as reference indices. Losses incurred as such may exceed the value of margin collateral, in which case margin calls may be triggered. In the event that both parties agree to enter a replacement (or termination) transaction, the interest rates received (paid) under the new arrangement may differ from those in the original arrangement, even if terms other than the interest rates are identical to those in the original transaction. Risks vary by transaction. Please thoroughly read the written materials provided, such as documents delivered before making a contract and disclosure statements.

In OTC transactions of credit default swaps (CDS), no sales commission will be charged. When entering into CDS transactions, the protection buyer will be required to pledge or entrust an agreed amount of margin collateral. In some of these cases, the transaction payments may exceed the amount of margin collateral. There shall be no advance notification of required collateral value or collateral ratios as they vary depending on the financial position of the protection buyer. CDS transactions carry the risk of losses owing to changes in the credit position of some or all of the referenced entities, and/or fluctuations of the interest rate market. The amount the protection buyer receives in the event that the CDS is triggered by a credit event may undercut the total amount of premiums that he/she has paid in the course of the transaction. Similarly, the amount the protection seller pays in the event of a credit event may exceed the total amount of premiums that he/she has received in the transaction. All other conditions being equal, the amount of premiums that the protection buyer pays and that received by the protection seller shall differ. In principle, CDS transactions will be limited to financial instruments business operators and qualified institutional investors. Transfers of equities to another securities company via the Japan Securities Depository Center are subject to a transfer fee of up to ¥11,000 (tax included) per issue transferred depending on volume. No account fee will be charged for marketable securities or monies deposited.

Nomura Securities Co., Ltd.

Financial instruments firm registered with the Kanto Local Finance Bureau (registration No. 142)

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